11b · Self-selected ad exposure, on real data — the Criteo experiment (IV · CausalPy)

July 11, 2026

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> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.** First run downloads the 311 MB Criteo file into `~/.cache/cmp` (one-time; a seeded subsample is cached after that).

The business decision. Notebook 11's story, at industrial scale and with nothing simulated: **13.98 million real users** from Criteo's advertising incrementality tests (Diemert et al. 2018). Users who *saw* an ad visit the advertiser at hugely elevated rates — but the ad system *chose* whom to show ads to (bids, browsing, auctions), so exposure is **self-selected** and the naive comparison credits the ad with intent that was already there. We want the true causal effect of exposure, to set the **budget cap per exposed user**.

1.0.1 Why this dataset is a gift for IV

Criteo's incrementality tests randomize **eligibility**: a random 15% of users are locked out of the campaign entirely. That randomized `treatment` flag is a textbook **encouragement instrument** for the self-selected `exposure`:

- **Relevance** — eligibility moves exposure (3.6% of eligible users end up exposed) — *testable*, first-stage F below;
- **Exclusion** — being *silently eligible* does nothing to your visiting behaviour unless an ad actually reaches you — *untestable*, stress-tested in Step 6;
- **Exogeneity** — eligibility is randomized by the test infrastructure — *holds by design*;
- **Monotonicity, for free**: ineligible users are **never** exposed (one-sided noncompliance), so "defiers" are impossible *by construction*, not by assumption — and the LATE specializes to the **effect on the exposed** (ATT), exactly the population a budget cap is about.

1.0.2 What replaces the simulator's planted truth

Two real-data anchors. **(a)** At n = 13.98M the design-based **Wald estimate** (intent-to-treat ÷ first stage) is essentially free of sampling noise — we compute it once on the full file and grade every subsample estimate against it. **(b)** The features let us *photograph the confounding itself*: f0–f11 are balanced across the randomized arms but sharply imbalanced across exposure — the self-selection nb11 could only simulate, measured.

```
FAST=False  subsample=200,000 rows  Bayesian fit on 100,000 rows
```

1.1 2 · Load the real data (what replaces "simulate a ground truth")

`cmp.data.load_criteo_uplift()` downloads the full file once, then caches a **seeded, balanced subsample** — `N_ARM` rows from each randomized arm. Balancing the 85/15 arms is legitimate *because* the sampling key is the randomized flag itself (both the ITT and the first stage are conditional on `treatment`, so selecting on it biases nothing) and roughly doubles precision per row. The **data card**:

fact	value
population	13,979,592 users, assembled from several incrementality tests
randomization	`treatment` = eligible to be targeted (85%) vs locked out (15%)
self-selection	`exposure` = actually saw an ad: **3.6% of eligible, 0.0% of ineligible**
outcomes	`visit` (4.7% base) — primary; `conversion` (0.29%) — secondary, rarer & noisier
features	f0–f11, anonymized dense floats (used for balance checks, not needed for IV)
full-data anchor	Wald LATE on visit **+28.7 pp**, on conversion **+3.2 pp** (computed below)

The one modelling accommodation, inherited from nb11 and disclosed the same way: outcome and exposure are 0/1, and CausalPy's joint-MvNormal IV treats both equations as **linear-probability** (Gaussian) — fine for the point estimate and the error correlation rho, so those are what we read.

```
[cmp.data] parsing the full 13.98M-row file to draw a balanced sample (~1 min)
```

```
subsample: 200,000 rows (100,000 eligible / 100,000 locked out)
P(exposed | eligible) = 0.0365   P(exposed | locked out) = 0.0000 -> one-sided
noncompliance, first-stage F = 3,793
visit rate: 0.0382 locked out vs 0.0493 eligible (ITT +1.11 pp)
full-data anchor (n=13.98M): ITT +1.034 pp, first stage 0.0360, Wald LATE +28.7
pp on visit
```

```
          f0         f1        f2        f3         f4        f5         f6  \
0  24.063211  10.059654  8.538476  4.679882  11.029585  3.013064  -5.116672
1  25.236557  10.059654  8.214383  4.679882  10.280525  4.115453 -13.583126
2  23.746553  10.059654  8.214383  4.679882  10.280525  4.115453 -10.764422
3  12.616364  10.059654  8.605682  4.679882  10.280525  4.115453   0.294443
4  25.194601  10.059654  8.214383  4.679882  10.280525  4.115453  -2.411115

          f7        f8         f9       f10       f11  treatment  conversion  \
0  11.299456  3.819627  29.642143  6.079819 -0.168679        0.0         0.0
1   4.833815  3.971858  13.190056  5.300375 -0.168679        1.0         0.0
2   4.833815  3.971858  13.190056  5.300375 -0.168679        0.0         0.0
3   4.833815  3.816414  34.180687  5.300375 -0.168679        1.0         0.0
4   4.833815  3.971858  13.190056  5.300375 -0.168679        1.0         0.0
```

```
   visit  exposure
0    0.0       0.0
1    0.0       0.0
2    0.0       0.0
3    0.0       0.0
4    0.0       0.0
```

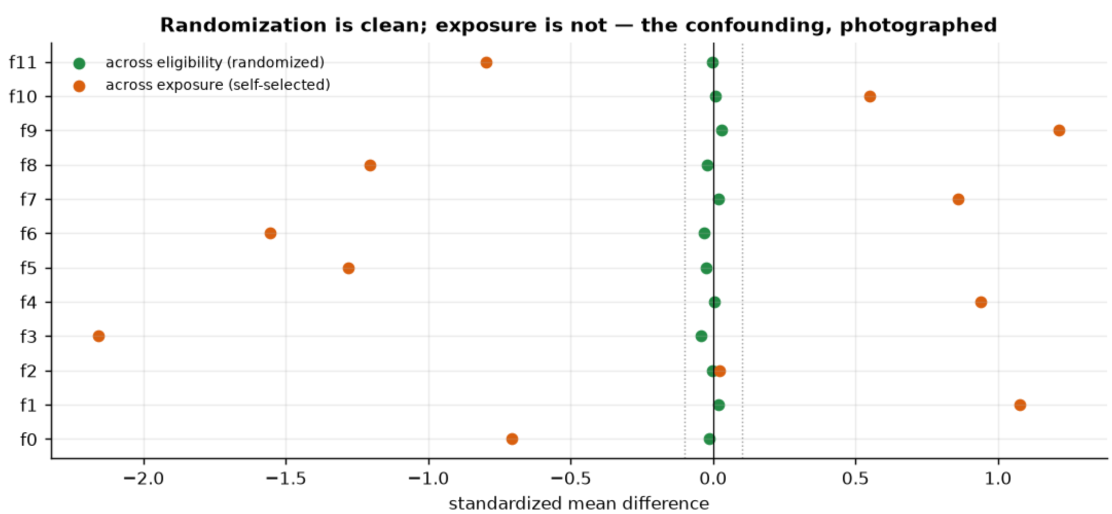
1.2 3 · Identify — a real instrument, and the confounding made visible

Exposure is endogenous: the ad system bids more for users who browse more, and browsing predicts visiting with or without ads. The instrument's three conditions are laid out in the header; here is the part a simulator can never show. If randomization worked, the twelve user features must be **flat across the arms** (any imbalance would impeach the instrument's exogeneity). If exposure is truly self-selected, the same features must be **skewed across exposure**. Both facts are checkable — one chart, two stories:

- across **eligibility** (randomized): standardized mean differences ~ 0 -> the instrument is clean;
- across **exposure** (chosen by the auction): large SMDs -> the exposed *were different people before any ad effect* — the very confounder nb11 called `engagement`, photographed in real data.

This is also why no regression on f0–f11 could rescue the naive estimate: these are only the *observable* shadows of targeting; the auction also used signals not in the file (the unmeasured confounder), which is precisely the case where notebook 05's adjustment toolkit surrenders and IV takes over.

```
max |SMD| across eligibility: 0.042  (randomization holds)
max |SMD| across exposure:    2.157  (the exposed were different people to begin
with)
```



1.3 4 · Estimate — OLS, the Wald ladder, and Bayesian IV

Same ladder as nb11, every rung now a real number:

- **OLS / naive**: visit rate of exposed minus unexposed — the number an attribution dashboard would report, inflated by targeting;
- **Reduced form** (eligibility -> visit) and **first stage** (eligibility -> exposure): both clean, both tiny — the encouragement is randomized but weak *per user*, which is why their **ratio** (Wald) is the honest per-exposure effect;
- **Bayesian IV** (CausalPy): the same ratio estimated jointly, with the endogeneity rho estimated rather than assumed away, and weakly-informative priors (nb11's gotcha: CausalPy's default 2SLS-centred `sigma=1` priors would produce an overconfident band). The fit uses `N_FIT` rows — NUTS cost scales with n, and Step 5 grades this subsample posterior against the 13.98M-row anchor to show what the smaller n costs us.

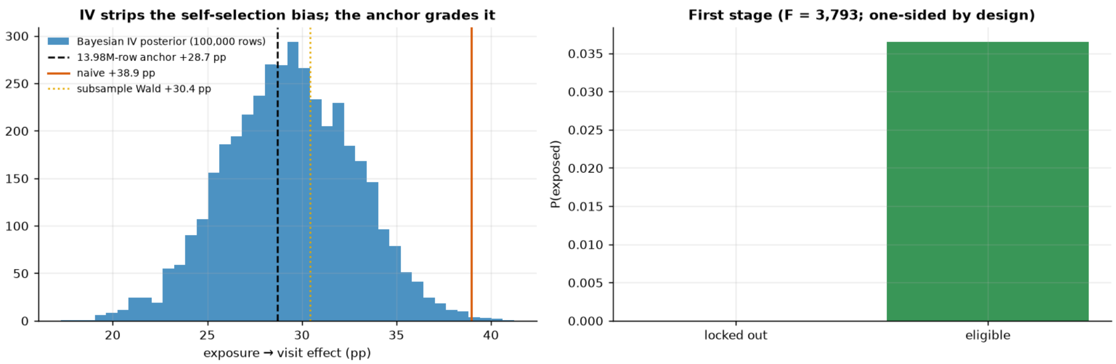
```
OLS / naive (exposed vs not)   +38.9 pp
Reduced form (Z->Y) +1.11 pp  ÷  first stage (Z->D) 0.0365  =  Wald +30.4 pp
Bayesian IV (100,000 rows)    +29.4 pp   90% credible interval [+23.8, +35.0] pp
full-data anchor (13.98M)      +28.7 pp
IV convergence: max r-hat 1.010 - min ESS 1201 - divergences 0
estimated endogeneity rho = +0.06 [90% +0.03, +0.10] - positive rho is the self-
selection: the same hidden leanings push a user into both exposure and visiting
```

1.4 5 · Validate — against 13.98 million rows

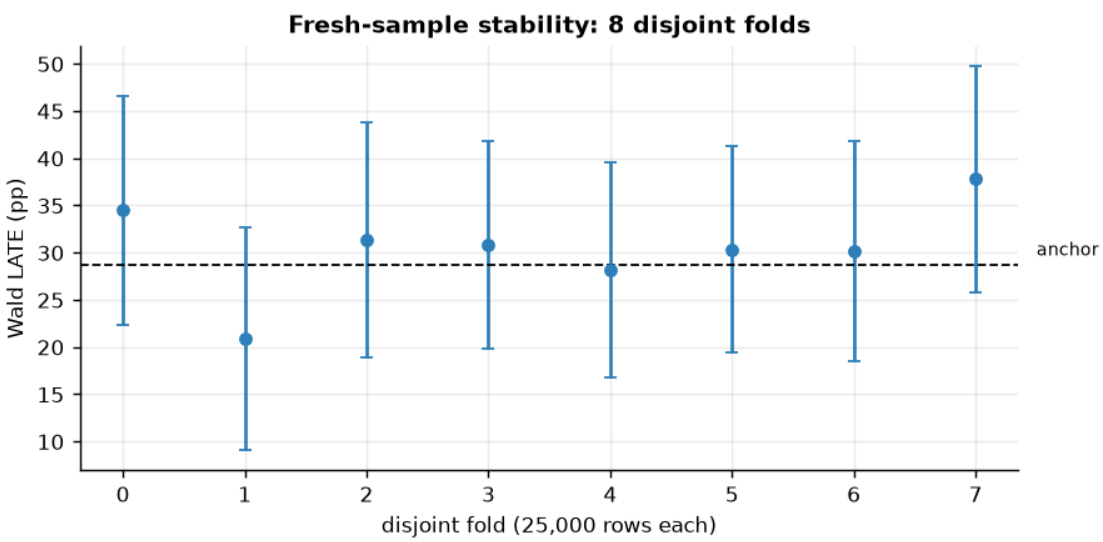
No planted truth, but a better yardstick than most real projects ever get: the full-file **Wald anchor**, whose own sampling noise at n = 13.98M is negligible for our purposes. Three checks, plus the star exhibit:

1. **Anchor recovery.** The subsample Bayesian IV's 90% interval should cover the anchor (+28.7 pp), and its point should sit within a couple of pp — that is the real-data analogue of nb11's "recover the €15."
2. **Internal consistency.** Wald ~ Bayesian IV on the same rows (they are the same identification strategy, estimated two ways); a gap would flag a modelling artefact, not a causal discovery.
3. **Disjoint-subsample stability** — nb11's multi-seed loop, real-data edition: we can't refit on fresh *worlds*, but 13.98M rows let us refit on fresh *samples*. Split the working subsample into disjoint folds, compute each fold's Wald +/- normal CI, and check spread and anchor coverage.
4. **The OLS − IV gap *is* the measured self-selection bias**: the euros an attribution dashboard would over-credit to the ad. Here it is ~8.5 pp on a ~29 pp effect — the dashboard number runs ~30% too big (32% on the full file), on 13.98 million real users.

```
anchor recovery: 90% interval [+23.8, +35.0] pp COVERS the 13.98M-row anchor
(+28.7 pp)
internal consistency: subsample Wald +30.4 vs Bayesian +29.4 pp
self-selection bias, measured: naive +38.9 -> causal +29.4 pp (+8.5 pp of pure
targeting, ~28% inflation)
```




```
fold Walds: mean +30.5 pp, sd 4.9 pp (anchor +28.7); 90% CIs cover the anchor in
100% of folds
the spread is the honest price of subsampling a rare exposure (3.6%) - and why
the anchor was computed on all 13.98M rows
```

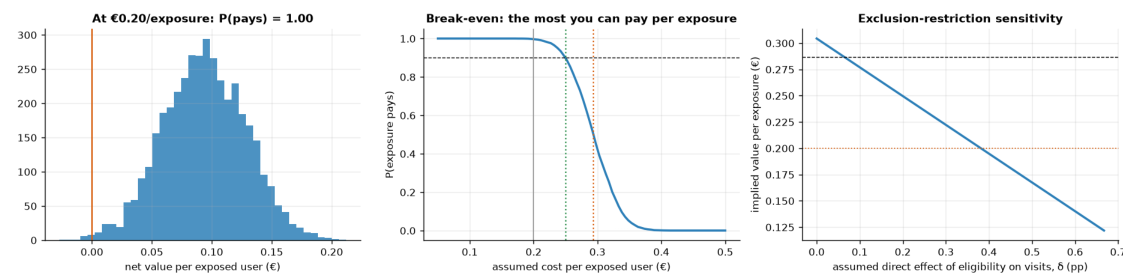


1.5 6 · Decide, in euros — the budget cap, and the exclusion stress test

The cap logic of nb11, with real numbers: an exposed user visits with **+29 pp** extra probability, so at an assumed **€1 value per incremental visit** an exposure is worth ~€0.29 — and the *naive* number would have priced it at ~€0.39. The decision rule: **BUY exposure while P(value x LATE > cost) >= 0.9**; sweeping the cost turns that into the **maximum viable price per exposed user**. Both business inputs are assumptions and get swept: the €1 visit-value scales the cap linearly, and the secondary `conversion` outcome gives an independent cap for a margin-per-conversion framing.

The exclusion stress test. The lockout is silent, but eligibility could still leak around the `exposure` flag — e.g. eligible users win *other* auctions differently, or "exposed" mislabels a below-the-fold impression. As in nb11: give eligibility a hypothetical direct effect delta on visits, correct the reduced form, and find the delta that would flip the BUY.

```
at €0.20/exposed user and €1.00/visit: net €+0.09, P(pays) = 1.00 -> BUY (to the
exposed margin)
budget cap: clears the 90% bar up to €0.25/exposed user; a coin flip at €0.29 -
and the naive number would have set the cap at €0.39, 33% too high
conversion framing (secondary): full-data LATE +3.20 pp -> at €40
margin/conversion an exposure is worth ~€1.28 - same BUY/HOLD shape, noisier
rung
exclusion stress: the BUY at €0.20 survives until eligibility's direct effect
exceeds delta ~ 0.39 pp - 35% of the whole reduced form (+1.11 pp) would have to
bypass exposure
```



1.6 7 · Caveats

- **ATT of the exposed, not everyone.** One-sided noncompliance makes monotonicity free, but the estimate speaks only for the ~3.6% of users the auction actually reached — the right population for "what is an exposure worth," the wrong one for "what if we doubled reach" (the next users the auction would reach are, by its own logic, worse prospects).
- **Exclusion is the load-bearing untestable.** The lockout is silent, but any eligibility side channel (auction dynamics on other campaigns, mislabelled below-the-fold "exposures") leaks into the estimate; step 6 priced the leak that would flip the call.
- **Linear-probability approximation** — binary outcome and binary exposure in a Gaussian joint model, as in nb11: read the point and rho, not the equation-level posterior widths.
- **Visits != profit.** The €1/visit and €40 margin/conversion are finance inputs, swept but assumed; the conversion-based rung is noisier (0.29% base rate) and its cap should be re-derived at decision time with the firm's own margin.
- **A composite of several tests** — Criteo assembled multiple incrementality tests; the LATE is a blend across campaigns/periods (fine for a benchmark cap, blur it into one campaign's plan with care).
- **Subsampling is disclosed, not hidden**: every number above is grade-able against the full-file anchor, and the fold check shows exactly how much noise a 200k-row view carries.

Cite the data: Diemert, Betlei, Renaudin & Amini (2018), *A Large Scale Benchmark for Uplift Modeling*, AdKDD. Dataset: Criteo AI Lab, research use.